

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY 2024

Program	: B.E. – Common to all Programs	Semester	: I / II
Course Name	: Design Thinking	Max. Marks	: 100
Course Code	: AECC17 / AECC27	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Assume missing data suitably.

UNIT - I

- What is design thinking? List and describe the different phases of design thinking. CO1 (08)
 - Discuss the importance of design thinking in business and education sectors. CO1 (08)
 - Differentiate between "Growth Mindset" and "Beginner's Mindset". CO1 (04)
- "A designer need to work with the full potential towards design". Describe any five key mindsets required to achieve it with suitable examples. CO1 (10)
 - Briefly discuss the importance of the 4C's namely critical thinking, creative thinking, communication and collaboration as an essential skill set for students with suitable examples CO1 (08)
 - Why is it called 'phases' of design thinking instead of 'stages'? CO1 (02)

UNIT - II

- Identify the target user market for the following brands, and explain briefly why you chose that group as the target user market. (i) Whiteboards, (ii) Wireless Computers keyboard (iii) Water spray can (iv) Nylon rope. CO2 (08)
 - What are the three elements of the problem statement? Give examples. CO2 (06)
 - Write ten interview questions to ask hospital staff pertaining to the issues regarding the disposal of gloves and needles. CO2 (06)
- What is Synthesis phase? What does one do during this phase? CO2 (06)
 - What is the importance of how might we' question? CO2 (06)
 - Identify the target user market for waterproof coating, and explain briefly why you chose that group. CO2 (08)

UNIT - III

- Discuss the myths pertaining to creativity and explain the related truth with suitable examples. CO3 (08)
 - What are the critical goals of building a prototype? Why designers should not get too attached to their prototypes? CO3 (04)
 - Differentiate between divergent thinking and convergent thinking with suitable examples. CO3 (04)
 - How can feedback be incorporated into the Ideate and Prototyping phases? CO3 (04)

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6. a) What is the objective of the Ideate phase in Design Thinking? Discuss the principles that were used in the IDEO and shopping cart challenge. CO3 (08)
b) What is Rapid Prototyping? how is it beneficial in design thinking. CO3 (04)
c) How can creativity, innovation and originality be achieved during Ideation phase of design thinking? CO3 (04)
d) Explain the role of facilitation and mentorship in the Prototyping phase. CO3 (04)

UNIT- IV

7. a) Assume the company you work for came with a unique design prototypes of two versions of smart watches and you are given the task to do usability test for them. What type of test would you choose and why? Explain. CO4 (08)
b) What exactly is user testing and why is it so important? CO4 (06)
c) What are some of the most common user testing methods used in Design Thinking? CO4 (06)
8. a) What is the end goal of testing stage in design thinking? CO4 (06)
b) Assume the company you work for came with a unique design prototype of a computer mouse and you are given the task to do usability test. What type of test would you choose and why? Explain. CO4 (08)
c) Choose any one most Common Pitfalls in Prototyping and suggest an appropriate way on how to avoid them. CO4 (06)

UNIT - V

9. a) "A healthcare facility wants to redesign its waiting room to improve patient experience and reduce anxiety". How would you use design thinking to involve patients and healthcare providers in the design process and create a welcoming and comfortable environment? CO5 (07)
b) "A non-profit organization is looking to tackle the issue of plastic pollution in oceans". How would you apply design thinking to identify innovative solutions to raise awareness, change behaviors, and reduce the amount of plastic waste entering the oceans? CO5 (07)
c) What are the key takeaways from the Prototype of Smart Village case study? CO5 (06)
10. a) "A restaurant owner wants to reduce food waste without compromising the quality of the food served". Identify and implement innovative solutions to address this challenge by applying all the phases of design thinking. CO5 (07)
b) "You are tasked with creating an online learning platform for a university to offer courses to remote students". How would you apply design thinking to design an intuitive and engaging platform that meets the diverse learning needs of students and educators? CO5 (07)
c) What were the challenges and solutions presented in Redesigning the Customer Contact Center at Toyota case study? CO5 (06)
